# Azim Ilkal

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## Education

**University of California, Davis** **Davis, CA**

B.S. Mechanical and Aerospace Engineering · Junior · GPA: 3.6 **Expected June 2028**

- Relevant Coursework: Fluids, Thermodynamics, Mechanics of Materials, Manufacturing Processes, MATLAB, FEA.

## Experience

**Birdstop** **March 2025 – Present**

*Flight Systems Intern (June 2026 – August 2026)* Detroit, MI

- Modeled quadcopter wind loading in Ansys Fluent (k-omega SST) across 20 wind angles and 4 design configurations.
- Applied local face sizing and inflation layers to resolve flow within a 1 million-cell limit.
- Built $F \propto V^2$ failure model using gauge-measured friction; 35 mph worst case, Reynolds-independent over 5–200 mph.
- Designed a 3D-printed shield for the new perch design that raised predicted critical wind speed from 35 to 88 mph.
- Validated the model with wind-blower testing of the new perch, confirming sliding as the failure mode at 80 mph.

*Field Engineer (March 2025 – Present)* Concord, CA

- Engineered 50+ FDM/SLA parts to consolidate drone and beacon electronics, shrinking beacon height from 4 to 3 ft.
- Documented designs with dimensioned, toleranced SolidWorks drawings to communicate changes with teammates.
- Reduced beacon build time from 3 days to 1 day via a redesigned layout for a 10-unit deployment to Spanish ATC.
- Flew 8+ Part 107 UAV tests; led 35–55°C thermal chamber tests isolating overheating to the battery and Jetson.

**National Association of Rocketry** **October 2025 – Present**

*Manufacturing Officer* Davis, CA

- Manufactured fins and nose cones for 2 rockets; built the avionics bay in Fusion 360 (GPS, altimeter, flight computer).
- Laid fiberglass on a nose cone for the competition's 2 L water payload that released the targeted 1 L at apogee.
- Placed 2nd of 17 teams at a FAR rocketry competition, reaching a GPS-measured 12,106 ft against a 12,243 ft target.

**Formula SAE** **January 2025 – December 2025**

*Composites Technician* Davis, CA

- Built the carbon-fiber front wing, rear wing, and endplates for the competition car; designed the jigs, machined molds on a CNC router and laser cutter, and laid up parts using wet layup and vacuum bagging.
- Contributed STAR-CCM+ drag/downforce simulations of 5+ wing designs for the team’s first aero package, which competed at Formula SAE Michigan.

**Unitrans** **September 2025 – Present**

*Shop Assistant (Bus Technician)* Davis, CA

- Diagnosed and repaired mechanical, hydraulic, pneumatic, structural, and electrical faults across a 50+ bus fleet, closing 25+ work orders monthly.
- Documented and reported mechanical defects in a work-order system to support compliance with DOT requirements.

## Projects

**Solar-Powered Weather Station** **December 2025 – February 2026**

- Designed a 3D-printed, weather-resistant ABS enclosure to house a solar-powered Raspberry Pi telemetry system.
- Configured 5 sensors (BME280, magnetic encoder, Hall-effect, UV, and rain) to capture environmental data.
- Programmed the Pi in Python to log all sensor readings, with results within 5–10% of local reference data.

## Skills

**CAD & Analysis:** Ansys Fluent and Mechanical (CFD, FEA), SolidWorks, STAR-CCM+, CATIA, Onshape, Fusion 360.

**Manufacturing:** 3D Printers (FDM & Resin), Laser Cutter, Band Saw, Bridgeport Mill, Soldering, CNC, GD&T.

**Programming and Data:** Python, MATLAB, Simulink, Arduino, Excel.

**Certifications:** FAA Part 107, NAR HPR Level 1.